Faculty of Computer Science and Mathematics

Lab Exercise 3

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Overview:

This in-class exercise extends Lab Exercise 3. You will apply block and inline elements, for page organization, HTML entities, HTML5 input types and attributes (placeholder, autofocus, required), <datalist> and the <output> element to solve a realistic problem.

There are two problems. Your lecturer will assign one problem to you, or you may be asked to complete both.

Learning Outcomes

1. Differentiate and correctly use block and inline elements.
2. Organize a web page into sections using elements.
3. Apply HTML entities to display reserved and special characters.
4. Implement HTML5 input types, input attributes and <datalist> in a form.
5. Use the <output> element to display a calculated result dynamically.

Best of luck!

Problem A: Campus Bakery Online Order Page

A small bakery on campus wants customers to place pre-orders online. Customers must be able to see the price list, fill in an order form, and see the total price instantly before submitting the order. Build the page using HTML only (no CSS). You may choose any bakery name, products and prices.

Instructions

Create a file named lab3_ExerA.html that contains the following:

1. Page Structure

- Use `<!DOCTYPE html>` and organize the whole page using five `<div>` elements with the ids header, nav, content, sidebar and footer.

2. Header and Navigation

- In the header, display the bakery name using `<h1>`, and use a `<span>` inside the `<h1>` for a short slogan.
- In the nav, create at least three inline links (`<a>`) separated by " | ".

3. Block and Inline Elements

- In the content section, write an introduction paragraph (`<p>`) that contains at least three different inline elements, e.g. `<strong>`, `<em>`, `<span>` and `<a>`.
- Use at least three different block elements in the page (e.g. `<h2>`, `<p>`, `<ul>`, `<div>`).

4. HTML Entities (Sidebar and Footer)

- In the sidebar, show a "Price List" using a list. Use at least four different entities in the page, for example:
  - `®` for the brand name (e.g. CampusBake®)
  - `<` or `>` (e.g. Orders > 10 pieces get a free drink)
  - ` `, `£` or `€`, and `©` in the footer

5. Order Form (HTML5 Form Elements)

Inside a `<div>` in the content section, create an order form that contains:

- Customer name: text input with placeholder, required and autofocus.
- Phone number (`type="tel"`) and e-mail (`type="email"`), both required.
- Pick-up date (`type="date"`) and pick-up time (`type="time"`).
- Flavour: an input linked to a `<datalist>` with at least 4 flavours.
- Icing colour (`type="color"`) and sweetness level using a range slider (`min="1"`, `max="5"`, `step="1"`).
- A submit button.

6. Price Calculator (<output>)

- Add two number inputs: Cupcakes (RM 3.00 each) and Cake Slices (RM 5.00 each), each with `min="0"` and `value="0"`.
- Use the form oninput attribute and an `<output>` element (with the correct name and for attributes) to show the total price automatically, e.g. 2 cupcakes + 1 slice = 11.

7. Comments

- Use comments (`<!-- ... -->`) to explain each section of your code.

Testing

- Click Submit with all fields empty. What happens? Then enter "myemail" in the e-mail field and submit again. Write your observations as a comment at the end of your file.
- Validate the file at https://validator.w3.org (Validate by File Upload).

Submission

Save the file as lab3_ExerA.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab3_ExerA.html and Screenshot of validation) through Github > Update your index.html (if you hosting your lab in Github pages)

Problem A: Campus bakery Order Online Page (Codes and Output)

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>CampusBake Online Order</title>
</head>
<body>
    <!-- 1. Header Section -->
    <div id="header">
        <h1>CampusBake&reg;
            <span>Freshly Baked, Happily Served!</span>
        </h1>
    </div>

    <!-- 2. Navigation Section -->
    <div id="nav">
        <a href="#home">Home</a> |
        <a href="#price-list">Price List</a> |
        <a href="#order-form">Order Now</a>
    </div>

    <!-- 3. Content Section -->
    <div id="content">

        <h2 id="home">Welcome to CampusBake!</h2>

        <p>
            Welcome to <strong>CampusBake</strong>, your favourite campus bakery.
            Enjoy our <em>freshly baked</em> cupcakes and cakes.
            We provide <span>delicious treats</span> at affordable prices.
            <a href="#order-form">Place your order here!</a>
        </p>

        <!-- 4. Order Form Section -->
        <div id="order-form">

            <h2>Online Order Form</h2>

            <form oninput="total.value = (Number(cupcakes.value) * 3) + (Number(cakeSlices.value) * 5)">

                <p>
                    Customer Name:
                    <input type="text" name="customerName"
                    placeholder="Enter your name" required autofocus>
                </p>

                <p>
                    Phone Number:
                    <input type="tel" name="phone" required>
                </p>

                <p>
                    E-mail:
                    <input type="email" name="email" required>
                </p>

                <p>
                    Pick-up Date:
                    <input type="date" name="pickupDate">
                </p>

                <p>
                    Pick-up Time:
                    <input type="time" name="pickupTime">
                </p>

                <p>
                    Flavour:
                    <input type="text" name="flavour" list="flavours">

                    <datalist id="flavours">
                        <option value="Chocolate">
                        <option value="Vanilla">
                        <option value="Red Velvet">
                        <option value="Strawberry">
                    </datalist>
                </p>

                <p>
                    Icing Colour:
                    <input type="color" name="icingColour">
                </p>
```

```
                <p>
                    Sweetness Level:
                    <input type="range" name="sweetness"
                    min="1" max="5" step="1">
                </p>

                <!-- 5. Price Calculator Section -->
                <h3>Price Calculator</h3>

                <p>
                    Cupcakes (RM 3.00 each):
                    <input type="number" name="cupcakes"
                    min="0" value="0">
                </p>

                <p>
                    Cake Slices (RM 5.00 each):
                    <input type="number" name="cakeSlices"
                    min="0" value="0">
                </p>

                <p>
                    Total Price: RM
                    <output name="total" for="cupcakes cakeSlices">0</output>
                </p>

                <!-- Submit Button -->
                <button type="submit">Submit Order</button>

            </form>
        </div>
    </div>

    <!-- 6. Sidebar Section -->
    <div id="sidebar">

        <h2 id="price-list">Price List</h2>

        <ul>
            <li>Chocolate Cupcake - RM 3.00</li>
            <li>Vanilla Cupcake - RM 3.00</li>
            <li>Red Velvet Cake Slice - RM 5.00</li>
            <li>Strawberry Cake Slice - RM 5.00</li>
        </ul>

        <p>Orders &gt; 10 pieces get a free drink!</p>

    </div>

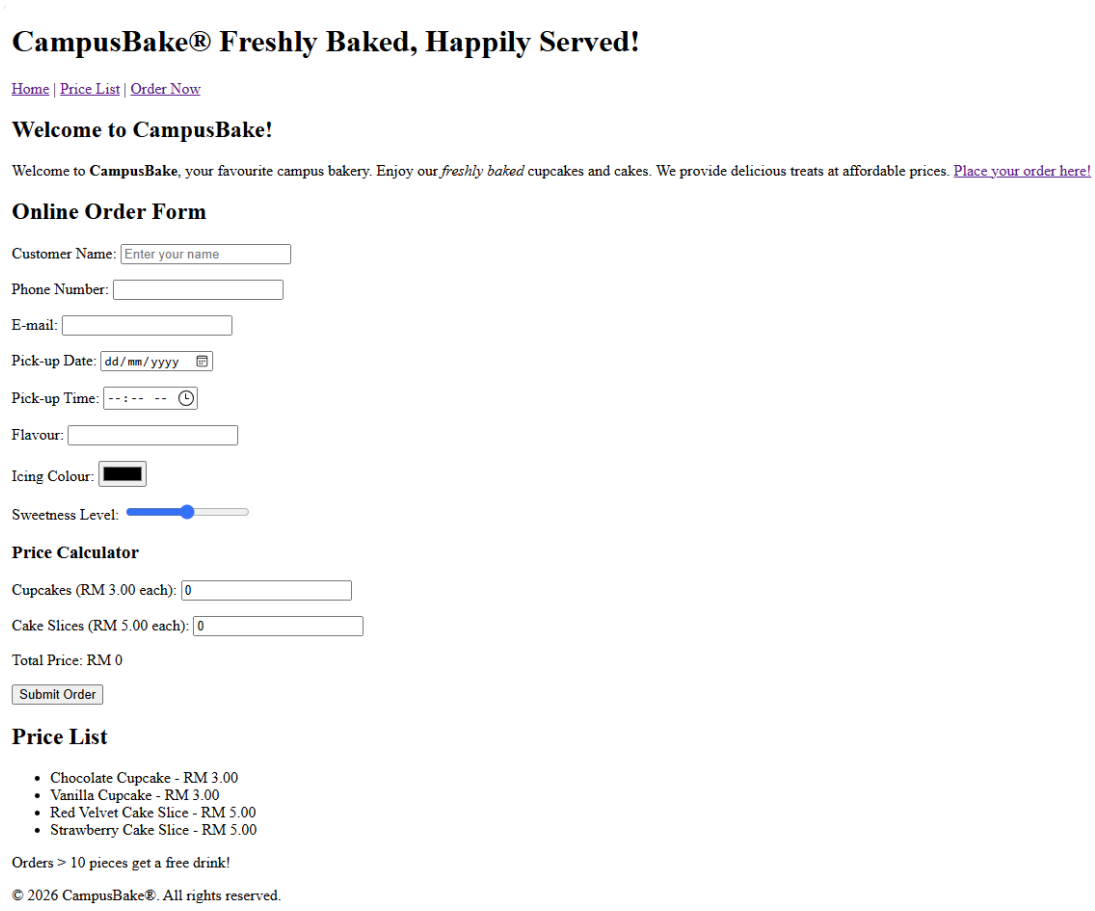
    <!-- 7. Footer Section -->
    <div id="footer">

        <p>&copy; 2026 CampusBake&reg;. All rights reserved.</p>

        <p>Thank you for supporting our bakery!</p>

    </div>
```

Output



CampusBake® Freshly Baked, Happily Served!

Home | Price List | Order Now

Welcome to CampusBake!

Welcome to **CampusBake**, your favourite campus bakery. Enjoy our *freshly baked* cupcakes and cakes. We provide delicious treats at affordable prices. Place your order here!

Online Order Form

Customer Name: Enter your name

Phone Number:

E-mail:

Pick-up Date: dd/mm/yyyy

Pick-up Time: --:-- --

Flavour:

Icing Colour:

Sweetness Level:

Price Calculator

Cupcakes (RM 3.00 each): 0

Cake Slices (RM 5.00 each): 0

Total Price: RM 0

Submit Order

Price List

- Chocolate Cupcake - RM 3.00
- Vanilla Cupcake - RM 3.00
- Red Velvet Cake Slice - RM 5.00
- Strawberry Cake Slice - RM 5.00

Orders > 10 pieces get a free drink!

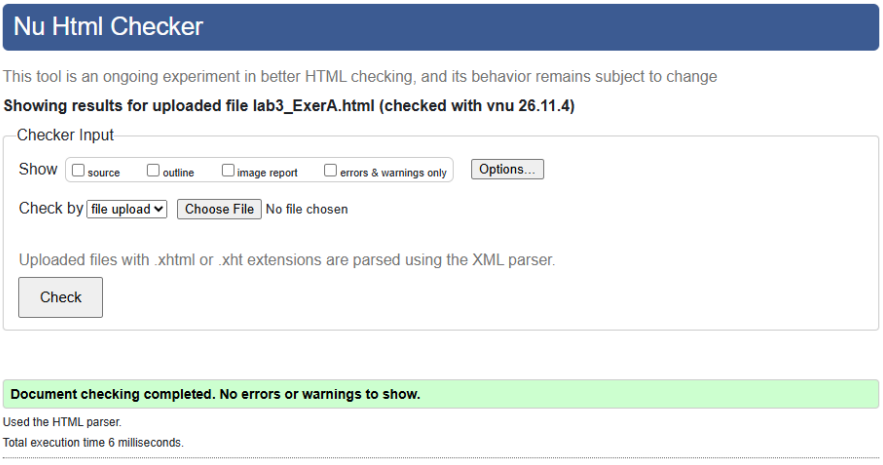
© 2026 CampusBake®. All rights reserved.

Testing

Click Submit with all fields empty. What happens? Then enter "myemail" in the e-mail field and submit again. Write your observations as a comment at the end of your file.

When clicking Submit with all fields empty,
the browser prevents submission because required fields are empty. When entering "myemail" in the email field, the browser prevents submission because the email format is invalid.

Validate the file at https://validator.w3.org (Validate by File Upload).



Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for uploaded file lab3_ExerA.html (checked with vnu 26.11.4)

Checker Input

Show source outline image report errors & warnings only Options...

Check by file upload Choose File No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Check

Document checking completed. No errors or warnings to show.

Used the HTML parser.

Total execution time 6 milliseconds.

Problem B: Ticket Booking (Debugging and Extend)

The campus cinema club built a ticket booking page, but users have reported several problems: the e-mail field accepts anything, the movie suggestions do not appear, the slider behaves strangely and the total shows "23" when the user books 2 adult and 3 child tickets. The copyright symbol also does not display. Find and fix the errors, then extend the page.

Given Code

```
 1  <!DOCTYPE>
 2  <html lang="en">
 3  <head>
 4      <meta charset="utf-8">
 5      <title>Ticket Booking</title>
 6  </head>
 7  <body>
 8  <div id="header">
 9      <h1>e-Ticket <span>Your Campus Cinema</h1></span>
10  </div>
11  <span id="content">
12      <h2>Book Your Ticket</h2>
13      <p>Promotion: tickets < RM10 for students &amp staff!</p>
14      <form action="booking.php" oninput="total.value = adult.value +
child.value">
15          <label for="name">Name:</label>
16          <input type="text" id="name" name="name" placeholder="Enter your
name" required autofocus><br>
17          <label for="email">E-mail:</label>
18          <input type="emial" id="email" name="email"><br>
19          <label for="date">Show date:</label>
20          <input type="date" id="date" name="date"><br>
21          <label for="movie">Movie:</label>
22          <input id="movie" name="movie" list="movies"><br>
23          <datalist id="movielist">
24              <option value="Space Rangers">
25              <option value="Kampung Comedy">
26          </datalist>
27          <label for="adult">Adult tickets:</label>
28          <input type="number" id="adult" name="adult" value="0"><br>
29          <label for="child">Child tickets:</label>
30          <input type="number" id="child" name="child" value="0"><br>
31          <label for="rating">Rate our service:</label>
32          <input type="range" id="rating" name="rating" min="5" max="1"><br>
33          Total tickets: <output name="total" for="adult kid"></output><br>
34          <input type="submit" value="Book Now">
35      </form>
36  </span>
37  <div id="footer">
38      <p>&Copy; 2026 CineKampus. All rights reserved.</p>
39  </div>
40  </body>
41  </html>54  </body>
55  </html>
```

Instructions

Part 1. Find and Fix the Errors

1. Copy the given code into a new file named lab3_ExerB.html.

2. Identify and correct all errors. At the top of the `<body>`, write an HTML comment listing each error and the line number, e.g. `<!-- Error 1 (Line 1): missing <!DOCTYPE html> -->`.

3. Test the page: the movie suggestions must appear, the e-mail field must reject "myemail", and 2 adult + 3 child tickets must show a total of 5.

4. Validate your corrected file at `https://validator.w3.org`. Your final page must show "No errors or warnings to show".

Part 2. Extend into a Two-Page Website

5. Add a `<div id="sidebar">` titled "Now Showing" that lists at least 3 movies. Each item must use inline elements: the title in `<strong>`, the age rating in `<span>` and a trailer link using <a>.

6. Add a phone number input (`type="tel"`) with a placeholder and required, and a show time input `(type="time").`

7. Add a second `<output name="price">` that shows the total price, where an adult ticket is RM 12 and a child ticket is RM 8 (e.g. 2 adults + 3 children = 48).

8. In the footer, add the text "e-Ticket® – Campus Cinema < RM10 every Tuesday" using the correct entities for ®, < and &.

9. Use comments to explain each new section of your code.

Submission

Save the file as lab3_ExerB.html, validate it at `https://validator.w3.org` (Validate by File Upload), and submit it (Submit lab3_ExerB.html and Screenshot of validation) through Github.

Problem B: Ticket Booking Debugging and Extend (Codes and Output)

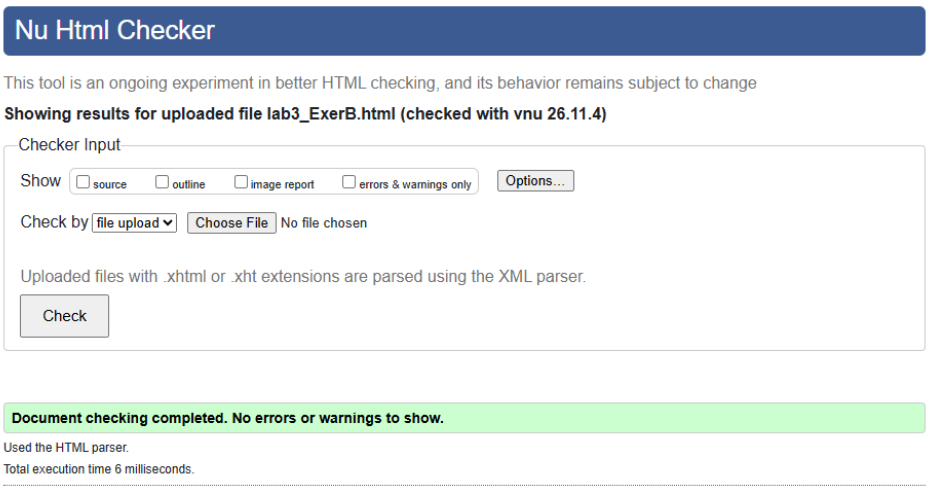
Error Identification

- Error 1 (Line 1): Incorrect DOCTYPE declaration. Changed to <!DOCTYPE html>.
- Error 2 (Line 9): Incorrect nesting of span and h1 tags. Corrected the order.
- Error 3 (Line 11): Used span for the main content. Changed to div.
- Error 4 (Line 13): Missing semicolon in & entity.
- Error 5 (Line 14): Incorrect addition of adult and child values as strings. Used Number().
- Error 6 (Line 18): Incorrect input type "emial". Changed to "email" and added required.
- Error 7 (Line 22): Datalist ID did not match the input list attribute.
- Error 8 (Line 23): Incorrect datalist ID. Changed to "movies".
- Error 9 (Line 32): Range min and max values were reversed. Corrected to min="1" max="5".
- Error 10 (Line 33): Incorrect output for attribute. Changed to for="adult child".
- Error 11 (Line 38): Incorrect copyright entity &Copy;. Changed to ©.
- Error 12 (Lines 54-55): Duplicate closing body and html tags removed.

```
1   <!DOCTYPE html>
2   <html lang="en">
3   <head>
4       <meta charset="utf-8">
5       <meta name="viewport" content="width=device-width, initial-scale=1.0">
6       <title>Ticket Booking</title>
7   </head>
8
9   <body>
10
11      <div id="header">
12          <h1>e-Ticket <span>Your Campus Cinema</span></h1>
13      </div>
14
15      <!-- Main content section -->
16      <div id="content">
17
18          <h2>Book Your Ticket</h2>
19
20          <p>Promotion: tickets &lt; RM10 for students &amp; staff!</p>
21
22          <!-- Ticket booking form -->
23          <form action="booking.php"
24                oninput="
25                total.value = Number(adult.value) + Number(child.value);
26                price.value = Number(adult.value) * 12 + Number(child.value) * 8;
27                ">
28
29              <label for="name">Name:</label>
30              <input type="text" id="name" name="name"
31                     placeholder="Enter your name" required autofocus>
32              <br>
33
34              <label for="email">E-mail:</label>
35              <input type="email" id="email" name="email"
36                     placeholder="Enter your email" required>
37              <br>
38
39              <!-- New section: Phone number input -->
40              <label for="phone">Phone Number:</label>
41              <input type="tel" id="phone" name="phone"
42                     placeholder="Enter your phone number" required>
43              <br>
44
45              <label for="date">Show date:</label>
46              <input type="date" id="date" name="date" required>
47              <br>
48
49              <!-- New section: Show time input -->
50              <label for="time">Show time:</label>
51              <input type="time" id="time" name="time" required>
52              <br>
53
54              <label for="movie">Movie:</label>
55              <input id="movie" name="movie" list="movies" required>
```

```
56
57              <!-- Movie suggestions -->
58              <datalist id="movies">
59                  <option value="Inside Out 2"></option>
60                  <option value="SPIDER-MAN: Brand New Day"></option>
61                  <option value="Toy Story 5"></option>
62              </datalist>
63              <br>
64
65              <label for="adult">Adult tickets:</label>
66              <input type="number" id="adult" name="adult"
67                     value="0" min="0" required>
68              <br>
69
70              <label for="child">Child tickets:</label>
71              <input type="number" id="child" name="child"
72                     value="0" min="0" required>
73              <br>
74
75              <label for="rating">Rate our service:</label>
76              <input type="range" id="rating" name="rating"
77                     min="1" max="5" value="3">
78              <br>
79
80              <!-- Ticket quantity output -->
81              Total tickets:
82              <output name="total" for="adult child">0</output>
83              <br>
84
85              <!-- New section: Total ticket price -->
86              Total price (RM):
87              <output name="price" for="adult child">0</output>
88              <br>
89
90              <input type="submit" value="Book Now">
91
92          </form>
93      </div>
94
95      <!-- New section: Sidebar showing available movies -->
96      <div id="sidebar">
97
98          <h2>Now Showing</h2>
99
100         <ul>
101             <li>
102                 <strong>Inside Out 2</strong>
103                 <span> (PG)</span>
104                 <a href="https://youtu.be/LEjhY15eCx0?si=B5jycLYVjOB3hPhC">
105                     Watch Trailer
106                 </a>
107             </li>
108
109             <li>
110                 <strong>SPIDER-MAN: Brand New Day</strong>
111                 <span> (PG-13)</span>
112                 <a href="https://youtu.be/62bIsvRcPv0?si=JJpIu2eQOWqQJ2qR">
113                     Watch Trailer
114                 </a>
115             </li>
116
117             <li>
118                 <strong>Toy Story 5</strong>
119                 <span> (PG)</span>
120                 <a href="https://youtu.be/c51ND9Hdbw0?si=cR6XkJ6vQM3weERD">
121                     Watch Trailer
122                 </a>
123             </li>
124         </ul>
125
126     </div>
127
128     <!-- Footer section -->
129     <div id="footer">
130         <p>e-Ticket&reg; &ndash; Campus Cinema &lt; RM10 every Tuesday &amp;copy; 2026 CineKampus. All rights reserved.</p>
131     </div>
132
133     </body>
134     </html>
```

Validate



Output

e-Ticket Your Campus Cinema

Book Your Ticket

Promotion: tickets < RM10 for students & staff!

Name: Enter your name

E-mail: Enter your email

Phone Number: Enter your phone number

Show date: dd/mm/yyyy

Show time: --:-- --

Movie:

Adult tickets: 0

Child tickets: 0

Rate our service:

Total tickets: 0

Total price (RM): 0

Book Now

Now Showing

- **Inside Out 2** (PG) Watch Trailer
- **SPIDER-MAN: Brand New Day** (PG-13) Watch Trailer
- **Toy Story 5** (PG) Watch Trailer

e-Ticket® – Campus Cinema < RM10 every Tuesday © 2026 CineKampus. All rights reserved.

Rubric

Table A: Rubric for Additional Lab Exercise 3 – Problem A

Criteria	Excellent (5)	Good (4)	Average (2–3)	Needs Improvement (0–1)
Page Structure	Uses all five sections (header, nav, content, sidebar, footer) with correct ids and logical organization.	Uses sections effectively with minor layout or naming issues.	Uses some elements but the structure lacks refinement or sections are missing.	Poor organization or does not use elements appropriately.
Block and Inline Elements	Correctly uses inside <h1>, inline links in nav, and at least 3 inline and 3 block elements with correct nesting.	Demonstrates block vs. inline elements with minor mistakes.	Attempts to use block and inline elements, but lacks clarity or nesting is incorrect.	Does not demonstrate understanding of block vs. inline elements.
HTML Entities	Correctly uses at least 4 different entities (incl. © and ®) in appropriate locations.	Uses entities with minor errors (e.g. only 3 types).	Attempts to use entities but with noticeable mistakes (wrong case, missing semicolon).	Fails to include or misuses HTML entities.
HTML5 Form Elements	Includes tel, email, date, time, color, range and datalist, with placeholder, required and autofocus used correctly.	Includes most required inputs and attributes with minor issues.	Includes some HTML5 inputs but lacks variety or attributes.	Lacks required form elements or has significant errors.
Use of <output> Element	Total price is calculated correctly and updates dynamically; <output> has correct name and for.	Implements <output> with minor issues (e.g. wrong for attribute).	Attempts <output> but calculation is incorrect (e.g. joins numbers as text).	Fails to implement or incorrectly uses <output>.
Total				**25 marks**

Table B: Rubric for Additional Lab Exercise 2 – Problem B

Criteria	Excellent (5)	Good (4)	Average (2–3)	Needs Improvement (0–1)
Error Identification	Identifies at least 8 errors correctly with line numbers and clear explanations.	Identifies 6–7 errors correctly.	Identifies 3–5 errors, or explanations are unclear.	Identifies fewer than 3 errors.

Criteria	Excellent (5)	Good (4)	Average (2–3)	Needs Improvement (0–1)
Error Correction & Validation	All errors corrected; page works as described and passes the W3C Validator with no errors.	Most errors corrected with 1–2 remaining validation errors.	Some errors corrected but the page still does not work correctly.	Few or no errors corrected.
Page Structure and Block/Inline Elements	Sidebar added with at least 3 movies using , and <a> correctly; content uses correctly.	Sidebar added with minor nesting or layout issues.	Sidebar attempted but inline elements are missing or misused.	Sidebar not added or block/inline elements misused.
HTML5 Form Elements and Entities	tel (with placeholder and required) and time inputs added; footer uses ®, < and & correctly.	Inputs and entities added with minor errors.	Some inputs or entities added with noticeable mistakes.	Inputs and entities not added or incorrect.
Use of <output> Element	Both outputs (total tickets and total price) calculate correctly and update dynamically.	Outputs work with minor issues.	Output attempted but calculation is incorrect.	Fails to implement the price output.
Total				**25 marks**